	Meaning
λ	Arrival rate
E[S]	Mean service time = E[S] = $1 / \mu$
$E[S^2]$	Second moment of service time
ρ	Utilisation
m	Number of servers

P-K formula

$$W = \frac{\lambda E[S^2]}{2(1-\rho)}$$

where $\boldsymbol{\rho} = \lambda / \mu$

Mean response time =
Mean waiting time + mean service time

(Week 1A)

Alice’s waiting time

= 5 + 6 + 3

Waiting time of an arriving customer

=

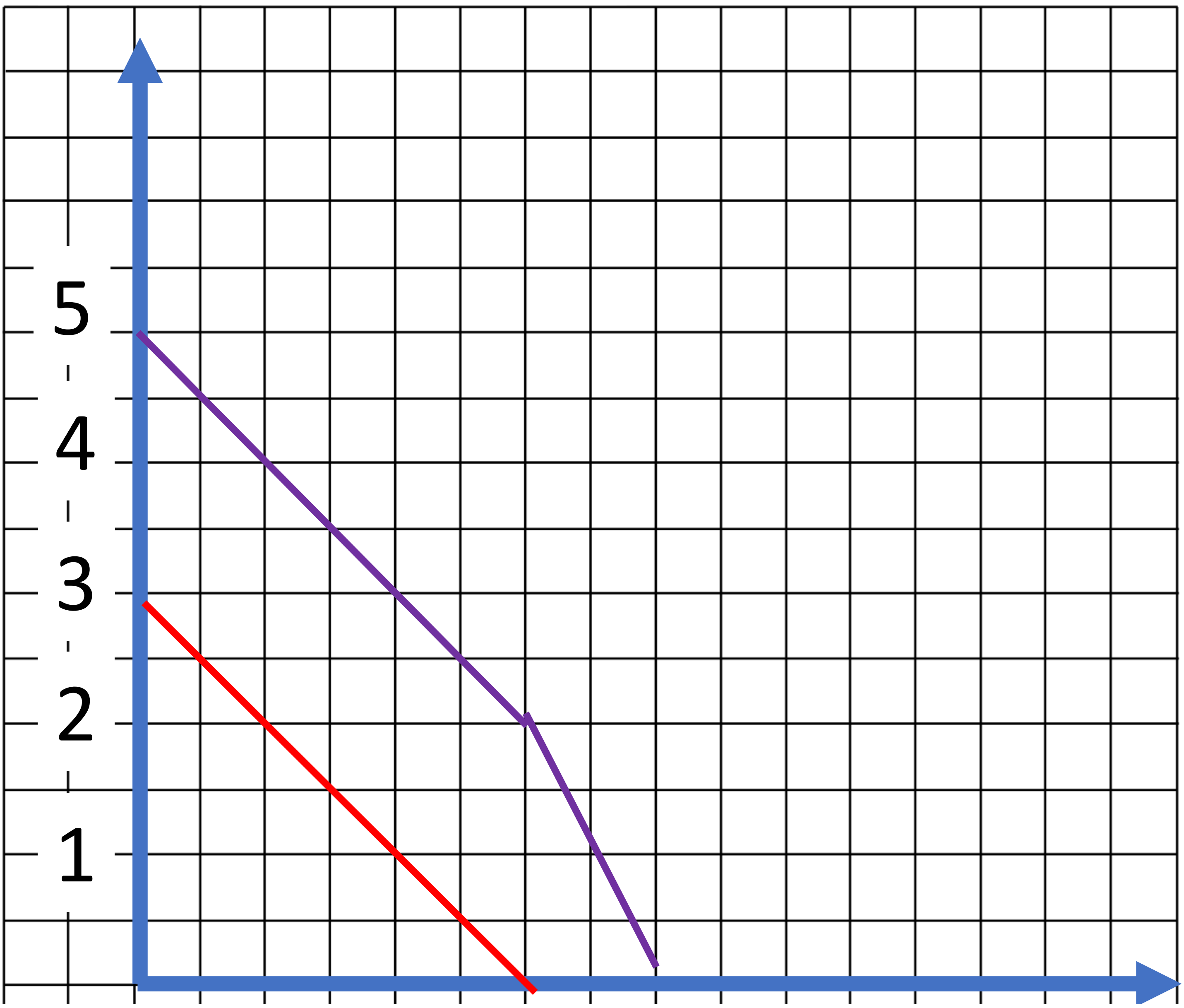
Time to serve the customers in the queue

+

Residual service time

Time t	Remaining work at time t	
	Job 1	Job 2
0	5	3
1	5-0.5 = 4.5	3-0.5= 2.5
2	4.5-0.5= 4	2.5-0.5=2
6	4-2=2	2-2=0
8	2-2=0	

Remaining work



Time t	Remaining work at time t			
	Job 1	Job 2	Job 3	Job 4
0	5	3		
1	5-0.5=4.5	3-0.5=2.5	4	
2	4.5-0.33 = 4.17	2.5-0.33 = 2.17	4-0.33 = 3.67	1
3	4.17-0.25 = 3.92	2.17-0.25 =	3.67-0.25 =	1-0.25 =
6